

SAOZ Spectrophotometer

Système d'Analyse par Observations Zénithales

FMI — Sodankylä Arctic Research Station · 67°22'N, 26°38'E

1. Instrument Overview

The SAOZ (**Système d'Analyse par Observations Zénithales**) is a UV–visible (300–650 nm) diode array spectrometer developed at the Service d'Aéronomie (now LATMOS, CNRS) in the late 1980s, following the discovery of the ozone hole by Farman et al. in 1985. It looks at sunlight scattered at zenith during twilight and uses Differential Optical Absorption Spectrometry (DOAS) to retrieve daily total columns of **ozone (O₃)** and **nitrogen dioxide (NO₂)** at sunrise and sunset, as well as total water vapour, oxygen dimer (O₄), and indicators for Polar Stratospheric Clouds (PSC) and volcanic aerosols.

A key advantage of the SAOZ is its use of the **Chappuis absorption bands of ozone (450–600 nm)** in the visible. Unlike UV-only instruments, this allows **continuous monitoring throughout the polar year**, in all weather conditions, even when UV signal is weak during low-sun seasons.

First deployed in 1988 simultaneously at **Dumont d'Urville** (Antarctica) and **Sodankylä** (Finland) (Pommereau and Goutail, 1988), the network has grown to over twenty instruments operated by CNRS and partner institutes at all latitudes, including the tropics. SAOZ data are used annually to monitor Arctic ozone depletion, for long-term trend analysis, and for validation of satellite O₃ and NO₂ retrievals (SBUV, TOMS, GOME, SCIAMACHY, OMI, GOME-2, IASI, OMPS-NPP, TROPOMI).

Key specifications

Spectral range: 300–650 nm (UV–visible)

Spectral resolution: ~1 nm (holographic grating)

Detector: Linear CCD, 2048 pixels

Measurement geometry: Zenith sky — scattered sunlight only

SZA window: 86°–91° (twilight: sunrise & sunset)

Measurement frequency: 2×/day (SR and SS)

Derived products: O₃, NO₂, H₂O, O₄ columns; PSC & volcanic aerosol detection

Accuracy (O₃): ~1–2% (column)

Temperature stability: ±0.1°C (thermoregulated housing)

2. Measurement Principle

2.1 Zenith scattered light geometry

Unlike the Brewer spectrophotometer, the SAOZ never points at the direct solar disk. Instead, it collects **diffuse skylight from the zenith**. At solar zenith angles (SZA) between 86° and 91°, the solar beam travels through an exceptionally long slant path in the atmosphere before being scattered downward. This greatly amplifies the integrated absorption signal of O₃ and NO₂ relative to daytime geometries, where SZA is smaller and the path through the ozone layer is shorter.

Measurements are therefore restricted to **two twilight windows per day**: one around **sunrise (SR)** and one around **sunset (SS)**. At the latitude of Sodankylä (67°N), the exact UTC times shift significantly across the year; in winter, the polar night limits or eliminates measurements entirely.

2.2 DOAS retrieval

Column amounts are retrieved using **DOAS — Differential Optical Absorption Spectroscopy**. The measured spectrum $I(\lambda)$ contains contributions from broadband scattering (Rayleigh, Mie) and narrow differential absorption features from O₃ and NO₂. The retrieval proceeds in three steps:

- The broadband background is removed by fitting and subtracting a low-order polynomial from $\ln(I(\lambda))$.
- The residual differential spectrum is fitted against reference laboratory absorption cross-sections of O₃ and NO₂.
- The fit coefficients, combined with the air mass factor (AMF) derived from the SZA, yield the total slant column, which is converted to a vertical column in Dobson Units (DU) for O₃ and in 10¹⁵ mol/cm² for NO₂.

3. Data Output — File Format & Contents

3.1 Accessing the data

SAOZ data for the Sodankylä station are publicly available from the CNRS/LATMOS server. Files are organized by year and station code:

```
Base URL : http://saoz.obs.uvsv.fr/saoz/  
File      : O3_<YYYY>.<STATION_CODE>  
Example   : http://saoz.obs.uvsv.fr/saoz/O3_2026.SK
```

Station codes:

SK = Sodankylä	NY = Ny-Alesund
SC = Scoresby-Sund	EU = Eureka
OH = OHP (France)	DD = Dumont d'Urville
DO = Dome C/Concordia	PA = Paris

Each file covers one full calendar year. A new file is generated daily and appended as measurements become available. If a year file is missing, the station was not operational.

3.2 Raw file structure

Each file is a plain ASCII text file with the following structure:

```
Line 1 : Units header
Line 2 : Column names
Lines 3+: Data rows (one row = one day)

Column names:
  Year  Month  Day  DoY  O3sr  O3ss  dO3sr  dO3ss  NO2sr  NO2ss  dNO2sr
dNO2ss
```

Column	Type	Unit	Description
Year	int	—	Calendar year
Month	int	—	Month (1–12)
Day	int	—	Day of month
DoY	int	—	Day of year (1–366)
O3sr	float	DU	O ₃ total column at sunrise
O3ss	float	DU	O ₃ total column at sunset
dO3sr	float	DU	1 σ uncertainty on O3sr
dO3ss	float	DU	1 σ uncertainty on O3ss
NO2sr	float	10 ¹⁵ mol/cm ²	NO ₂ column at sunrise
NO2ss	float	10 ¹⁵ mol/cm ²	NO ₂ column at sunset
dNO2sr	float	10 ¹⁵ mol/cm ²	1 σ uncertainty on NO2sr
dNO2ss	float	10 ¹⁵ mol/cm ²	1 σ uncertainty on NO2ss

Missing values are encoded as **NaN** in both column value and uncertainty fields. A **NaN** typically indicates cloud cover during the twilight window, instrument maintenance, or polar night (no twilight at the required SZA range).

3.3 Processed CSV output (download script)

The Python download script ([saoz_download.py](#)) retrieves the raw annual file, parses it, filters the requested date range, and writes a clean **.csv** to disk. The output CSV is identical in structure to the raw columns, with **NaN** replaced by empty cells for compatibility with pandas, R, and spreadsheet tools.

```
Output path: ./saoz_data/<STATION>/saoz_<STATION>_<START>_<END>.csv
Example:     ./saoz_data/SK/saoz_SK_2026-04-01_2026-04-30.csv
```

```
CSV columns (same as raw):
year, month, day, doy,
O3sr, O3ss, dO3sr, dO3ss,
NO2sr, NO2ss, dNO2sr, dNO2ss
```

4. Using the Download Script

4.1 Configuration

Edit the CONFIG block at the top of `saoz_download.py`:

```
STATION      = "SK"           # Station code (see table above)
DATE_START   = "2026-01-01"   # ISO format YYYY-MM-DD
DATE_END     = "2026-04-30"   # ISO format YYYY-MM-DD
OUT_DIR      = Path("./saoz_data") # Output directory (created if absent)
```

4.2 Running

```
pip install requests
python saoz_download.py
```

4.3 What the script does

- Computes the list of calendar years that intersect the requested date range.
- Downloads one raw file per year from the LATMOS server.
- Saves the raw file to disk as a local cache (e.g. O3_2026.SK).
- Parses every data row and converts NaN strings to Python None.
- Filters rows to keep only those within [DATE_START, DATE_END].
- Writes a single merged CSV spanning the full period.
- Prints a summary: number of valid days, O₃ and NO₂ ranges and means.

5. Scientific Context & Known Limitations

5.1 Strengths

- **Operational since the early 1990s at many NDACC stations. Sodankylä has >30 years of continuous data — invaluable for ozone trend detection.:** Long time series
- **Fully autonomous operation; no manual intervention needed at twilight. Ideal for polar sites with extreme weather.:** Robustness
- **O₃ and NO₂ retrieved simultaneously from the same spectrum.:** Dual species
- **Part of NDACC, SAOZ global network (~30 stations), and directly used for HEGIFTOM/TOAR-II trend analyses.:** Network integration

- **SAOZ performance has been evaluated in six major NDACC UV intercomparison campaigns: New Zealand (1992), Camborne UK (1995), Observatoire de Haute-Provence (1996), Andoya Norway (2003), Cabauw Netherlands (2009 and 2016). Results consistently confirm its reliability for long-term monitoring.:**
Independently validated

5.2 Limitations

- **No intra-day variability is captured. Pandora, Brewer, or TROPOMI are needed for sub-daily monitoring.:** Only 2 measurements per day
- **Only the total column is retrieved. BTS DIAL lidar or ECC ozonesondes are needed for altitude-resolved information.:** No vertical profile
- **Optically thick clouds during the twilight window cause missing values (NaN). Cloud fraction at SZA >86° can be high in Arctic spring.:** Cloud sensitivity
- **At 67°N, measurements are limited or absent from late November to mid-January when the sun does not reach SZA = 91°.:** Polar night gap
- **Zenith-scattered geometry requires careful modelling of the air mass factor; systematic errors at extreme SZA are on the order of 1–2%.:** Absolute calibration

6. Networks & Data Policy

Data are freely available from <http://saoz.obs.uvsq.fr/>. Users are requested to cite the SAOZ team and the relevant station PI (Mustapha Meftah, LATMOS) in any publication using these data.